Magnetic extensions of the self-consistent density-matrix framework for graphene islands

Goal

We start from the already implemented density-matrix framework with self-consistent Coulomb potential and extend it in consistent steps to include magnetic effects generated by optical excitation in graphene islands.[1, 2]

Step 1: Validate the electrostatic baseline (L0)

Step 2: Add the induced magnetic field in the Zeeman part of the Hamiltonian (L1)

Step 3: Add the induced vector potential as a Peierls phase (L2)

1 Background theory

The starting point is the single-particle density-matrix equation

$$i\hbar\dot{\rho}(t) = [H(t), \rho(t)] - \frac{i\hbar}{2\tau}\left(\rho(t) - \rho^{(0)}\right), \tag{1}$$

where $\rho^{(0)}$ is the equilibrium density matrix and τ is the phenomenological relaxation time.[1, 3]

For a finite graphene island, work directly in a real-space orbital basis $\{|l, \sigma\rangle\}$, where σ denotes the spin. The Hamiltonian is split as

$$H(t) = H^{\mathrm{TB}} + H_\phi[\rho](t) + H_Z[B_{\mathrm{ind}}](t) + H_{\mathrm{orb}}[A_{\mathrm{ind}}](t). \tag{2}$$

Here:

- H^{TB} is the ground-state graphene tight-binding Hamiltonian
- $H_\phi[\rho]$ contains the external optical field in length gauge and the self-consistent Hartree term
- $H_Z[B_{\mathrm{ind}}]$ is the induced Zeeman correction
- $H_{\mathrm{orb}}[A_{\mathrm{ind}}]$ is an optional induced orbital feedback term, which updates the Peirls phase based on the induced current.

2 Electrostatic part

The external optical excitation is represented *only* through the scalar potential.[1, 2] Keep the electrostatic self-consistency,

$$H_\phi[\rho](t) = e\sum_{l,\sigma} \phi_l(t)\, \hat{a}^\dagger_{l\sigma}\hat{a}_{l\sigma}, \tag{3}$$

with

$$\phi_l(t) = \phi_l^{\mathrm{ext}}(t) + \sum_{l'} v_{ll'}\, \rho_{l'}^{\mathrm{ind}}(t), \qquad \rho_l^{\mathrm{ind}}(t) = -e\sum_\sigma \left[\rho_{ll,\sigma}(t) - \rho^{(0)}_{ll,\sigma}\right]. \tag{4}$$

Here $v_{ll'}$ is the Coulomb matrix used already in the electrostatic implementation.

For an external optical field $\mathbf{E}_{\text{ext}}(t)$ in the dipole approximation,

$$\phi_l^{\text{ext}}(t) = -\mathbf{r}_l \cdot \mathbf{E}_{\text{ext}}(t). \tag{5}$$

The scalar potential therefore remains responsible for both the external optical field through ϕ_l^{ext} and the Coulomb response through the induced Hartree term.

Since there is no externally applied magnetic field, the external optical field is treated entirely in length gauge through Eq. (5). There is no external vector potential $\mathbf{A}_{\text{ext}}$ and no external magnetic field $\mathbf{B}_{\text{ext}}$ as these would double count the effect of the optical field.[1, 4]

3 Orbital magnetic coupling

The tight-binding Hamiltonian is

$$H^{\text{TB}} = \sum_{ll',\sigma} H_{ll'}^{\text{TB}}\, \hat{a}_{l\sigma}^{\dagger} \hat{a}_{l'\sigma}. \tag{6}$$

Only when the induced orbital magnetic feedback is included, does one introduce a Peierls substitution through an induced vector potential

$$H_{\text{orb}}[A_{\text{ind}}](t) = \sum_{ll',\sigma} H_{ll'}^{\text{TB}}\, e^{i\theta_{ll'}^{\text{ind}}(t)} \hat{a}_{l\sigma}^{\dagger} \hat{a}_{l'\sigma}, \qquad \theta_{ll'}^{\text{ind}}(t) = \frac{e}{\hbar} \int_{\mathbf{r}_{l'}}^{\mathbf{r}_l} \mathbf{A}_{\text{ind}}(\mathbf{r}, t) \cdot d\mathbf{r}. \tag{7}$$

For nearest-neighbor graphene, we have $H_{ll'}^{\text{TB}} = -t$ for connected sites and 0 otherwise. You can approximate the line integral by the midpoint rule,

$$\theta_{ll'}^{\text{ind}}(t) \approx \frac{e}{\hbar}\, \mathbf{A}_{\text{ind}}\left(\frac{\mathbf{r}_l + \mathbf{r}_{l'}}{2}, t\right) \cdot (\mathbf{r}_l - \mathbf{r}_{l'}). \tag{8}$$

4 Spin coupling

The Zeeman term depends on the induced magnetic field

$$H_Z[B_{\text{ind}}](t) = \frac{\mu_B}{2} \sum_l \begin{pmatrix} \hat{a}_{l\uparrow}^{\dagger} & \hat{a}_{l\downarrow}^{\dagger} \end{pmatrix} \left[g_l\, \mathbf{B}_{\text{ind}}(\mathbf{r}_l, t) \cdot \boldsymbol{\sigma}\right] \begin{pmatrix} \hat{a}_{l\uparrow} \\ \hat{a}_{l\downarrow} \end{pmatrix}. \tag{9}$$

Use isotropic $g_l = 2$. For a dominant out-of-plane induced field this becomes

$$H_Z(t) = \mu_B \sum_l B_{\text{ind},z}(\mathbf{r}_l, t) \left(\hat{a}_{l\uparrow}^{\dagger} \hat{a}_{l\uparrow} - \hat{a}_{l\downarrow}^{\dagger} \hat{a}_{l\downarrow}\right). \tag{10}$$

Spin-resolved extension of the Coulomb interaction

The Coulomb matrix elements already implemented from Clementi-type $2p_z$ orbital integrals should be kept spin independent. The reason is that the Coulomb interaction couples to charge density, not directly to spin. Spin enters only through the fermionic occupation operators and through the spin-resolved density matrix.[5]

In the original spin-degenerate implementation, the induced charge is written with an explicit factor of 2,

$$\rho_l^{\text{ind}}(t) = -2e \left[\rho_{ll}(t) - \rho_{ll}^{(0)}\right], \tag{11}$$

and the self-consistent scalar potential is

$$\phi_l(t) = \phi_l^{\text{ext}}(t) + \sum_{l'} v_{ll'}\, \rho_{l'}^{\text{ind}}(t), \tag{12}$$

where $v_{ll'}$ is the pairwise Coulomb interaction between carbon orbitals.

For a spin-resolved graphene model, the factor of 2 must be removed and replaced by an explicit sum over spin:

$$\rho_l^{\text{ind}}(t) = -e \sum_{\sigma=\uparrow,\downarrow} \left[\rho_{ll,\sigma}(t) - \rho_{ll,\sigma}^{(0)}\right]. \tag{13}$$

Thus, the induced charge is the *total* induced charge from both spin channels.

The self-consistent Hartree potential is then unaffected and acts identically on both spin species, and the density matrix is block diagonal in spin,

$$\rho(t) = \begin{pmatrix} \rho^{\uparrow}(t) & 0 \\ 0 & \rho^{\downarrow}(t) \end{pmatrix}. \tag{14}$$

5 Current operator

The bond current is obtained from the Hamiltonian derivative with respect to the Peierls phase.[4]

$$\hat{J}_{ll'}(t) = -\frac{1}{\hbar}\frac{\partial H}{\partial \theta_{ll'}}. \tag{15}$$

This gives the expectation value

$$J_{ll'}(t) = \frac{ie}{\hbar}\sum_{\sigma}\left[H_{ll'}^{\text{TB}} e^{i\theta_{ll'}(t)}\rho_{l'l,\sigma}(t) - H_{l'l}^{\text{TB}} e^{i\theta_{l'l}(t)}\rho_{ll',\sigma}(t)\right]. \tag{16}$$

For the ground-state tight-binding model one sets $\theta_{ll'}(t) = 0$, while for the model including the induced vector potential, one uses the induced phase from Eq. (7). This formulation is tied directly to the Hamiltonian and is therefore the safest way to preserve consistency between the propagated state and the current used to generate induced magnetic corrections.[4]

Bond-current geometry

For each connected pair of sites l, l', define:

$$\mathbf{r}_{ll'}^{c} = \frac{\mathbf{r}_l + \mathbf{r}_{l'}}{2}, \qquad \boldsymbol{\ell}_{ll'} = \mathbf{r}_l - \mathbf{r}_{l'}. \tag{17}$$

Here $\mathbf{r}_{ll'}^{c}$ is the midpoint of the bond and $\boldsymbol{\ell}_{ll'}$ is the bond vector.

The current $J_{ll'}(t)$ from Eq. (16) is interpreted as a current flowing along the bond direction $\boldsymbol{\ell}_{ll'}$. Associate to each bond the vector current element

$$\mathbf{I}_{ll'}(t) = J_{ll'}(t)\,\hat{\boldsymbol{\ell}}_{ll'}, \qquad \hat{\boldsymbol{\ell}}_{ll'} = \frac{\boldsymbol{\ell}_{ll'}}{|\boldsymbol{\ell}_{ll'}|}. \tag{18}$$

Only sum each bond once, for example with $l < l'$, to avoid double counting.

6 Induced magnetic field

For a planar graphene island in the xy plane, the most practical implementation is to compute the induced magnetic field directly from a discrete Biot–Savart law.

At site m, define

$$\mathbf{B}_{\text{ind}}(\mathbf{r}_m, t) \approx \frac{\mu_0}{4\pi}\sum_{\langle ll'\rangle} J_{ll'}(t)\,\frac{\boldsymbol{\ell}_{ll'} \times \left(\mathbf{r}_m - \mathbf{r}_{ll'}^{c}\right)}{\left|\mathbf{r}_m - \mathbf{r}_{ll'}^{c}\right|^3}, \tag{19}$$

Each bond current is treated as a short straight current segment centered at $\mathbf{r}_{ll'}^{c}$. Equation (19) is the discrete magnetostatic approximation appropriate for the quasistatic regime assumed throughout the project.

Updating the Zeeman term

Once Eq. (19) is available, the Zeeman term is updated locally at each site:

$$H_Z(t) = \frac{\mu_B}{2} \sum_m \begin{pmatrix} \hat{a}^\dagger_{m\uparrow} & \hat{a}^\dagger_{m\downarrow} \end{pmatrix} \left[g_m \, \mathbf{B}_{\text{ind}}(\mathbf{r}_m, t) \cdot \boldsymbol{\sigma} \right] \begin{pmatrix} \hat{a}_{m\uparrow} \\ \hat{a}_{m\downarrow} \end{pmatrix}. \tag{20}$$

For a dominant out-of-plane field this reduces to

$$H_Z(t) = \mu_B \sum_m B_z^{\text{ind}}(\mathbf{r}_m, t) \left(\hat{a}^\dagger_{m\uparrow} \hat{a}_{m\uparrow} - \hat{a}^\dagger_{m\downarrow} \hat{a}_{m\downarrow} \right). \tag{21}$$

7 Level 1 (L1) Implementation

This concludes the first level of implementing the magnetic effects (L1) which should be compared to the original purely electrostatic implementation. We will use the lattice bond currents $J_{ll'}(t)$ and construct a discrete induced vector potential and magnetic field and use this to update the Zeeman part of the Hamiltonian only,

$$H^{(L1)}(t) = \sum_{ll',\sigma} H^{\text{TB}}_{ll'} \hat{a}^\dagger_{l\sigma} \hat{a}_{l'\sigma} + e \sum_{l,\sigma} \phi_l[\rho](t) \hat{a}^\dagger_{l\sigma} \hat{a}_{l\sigma} + \mu_B \sum_l B_z^{\text{ind}}(\mathbf{r}_l, t) \left(\hat{a}^\dagger_{l\uparrow} \hat{a}_{l\uparrow} - \hat{a}^\dagger_{l\downarrow} \hat{a}_{l\downarrow} \right), \tag{22}$$

with

$$\phi_l[\rho](t) = -\mathbf{r}_l \cdot \mathbf{E}_{\text{ext}}(t) + \sum_{l'} v_{ll'} \, \rho^{\text{ind}}_{l'}(t). \tag{23}$$

The practical implementation consists of the following substeps,

1. compute bond currents $J_{ll'}(t)$ from the density matrix
2. convert bond currents into current elements located at bond midpoints
3. sum the magnetic contribution from all bond currents at each site
4. use the resulting induced field to update the Zeeman term

At each accepted time-step from t_n to t_{n+1}:

1. Predictor step: propagate ρ_n from t_n to t_{n+1} with RK45 using

   $$H^{(0)}(t, \rho) = H^{\text{TB}} + H_\phi[\rho, t] + H_Z[B^{(0)}_{\text{ind}}],$$

   where $B^{(0)}_{\text{ind}}$ is either zero (first step) or the accepted value from the previous step. Note that the Peierls phase is zero for this implementation level. This gives a trial density ρ^*_{n+1}.
2. Compute trial bond currents $J^*_{ll'}(t_{n+1})$ from ρ^*_{n+1}.
3. Compute the trial induced magnetic field $\mathbf{B}^*_{\text{ind}}(t_{n+1})$ from the Biot–Savart formula.
4. Corrector step: propagate again from the original state ρ_n over the same interval $[t_n, t_{n+1}]$ with RK45 using

   $$H^{(\text{corr})}(t, \rho) = H^{\text{TB}} + H_\phi[\rho, t] + H_Z[\mathbf{B}^*_{\text{ind}}(t_{n+1})].$$

   This gives the corrected density ρ^{corr}_{n+1}.
5. Accept

   $$\rho_{n+1} = \rho^{\text{corr}}_{n+1}.$$
6. Move to the next time-step.

8 Induced vector potential

To feed the induced field back into the Peierls phases, an induced vector potential is needed. The simplest discrete choice is

$$\mathbf{A}_{\text{ind}}(\mathbf{r}_m, t) \approx \frac{\mu_0}{4\pi} \sum_{\langle ll' \rangle} J_{ll'}(t) \, \frac{\boldsymbol{\ell}_{ll'}}{\left|\mathbf{r}_m - \mathbf{r}^c_{ll'}\right|}. \tag{24}$$

This is only an approximate lattice representation and should be regarded as a *model choice.*

Updating the Peierls phases

Update the induced Peierls phases bond by bond:

$$\theta^{\text{ind}}_{ll'}(t) = \frac{e}{\hbar} \int_{\mathbf{r}_{l'}}^{\mathbf{r}_l} \mathbf{A}_{\text{ind}}(\mathbf{r}, t) \cdot d\mathbf{r}. \tag{25}$$

Using the midpoint approximation,

$$\theta^{\text{ind}}_{ll'}(t) \approx \frac{e}{\hbar} \mathbf{A}_{\text{ind}}(\mathbf{r}^c_{ll'}, t) \cdot \boldsymbol{\ell}_{ll'}. \tag{26}$$

Important comment on gauge consistency. Equation (24) is *not* obtained from a controlled gauge-fixed solution of the electromagnetic field equations. It is a discrete Biot–Savart-inspired reconstruction of $\mathbf{A}_{\text{ind}}$ from the current and may therefore introduce gauge inconsistency when used inside the Peierls phase. In particular, two different approximate reconstructions of $\mathbf{A}_{\text{ind}}$ can yield similar $\mathbf{B}_{\text{ind}} = \nabla \times \mathbf{A}_{\text{ind}}$ but different bond phases $\theta^{\text{ind}}_{ll'}$. A note on gauge invariance is included in the final part of the note. A way to test the influence of this inconsistency, is to compare the directly computed field from Eq. (19) with the discrete curl of the reconstructed vector potential:

$$\mathbf{B}^{(\nabla \times A)}_{\text{ind}}(\mathbf{r}_m, t) = \nabla \times \mathbf{A}_{\text{ind}}(\mathbf{r}_m, t). \tag{27}$$

The important comparison will be how much these two vary compared to the size of the induced magnetic field itself.[4]

9 Level 2 (L2) implementation

$$H^{(\text{L2})}(t) = H_{\text{orb}}[A_{\text{ind}}](t) + H_\phi[\rho](t) + H_Z[B_{\text{ind}}](t), \tag{28}$$

where the induced vector potential is included in addition to the induced Zeeman term and combined with the density-matrix equation (1) and the current formula (16).

The self-consistent iteration within each time step is

$$\rho^{(n)} \to \rho^{(n)}_{\text{ind}} \to \phi^{(n)} \to J^{(n)} \to A^{(n)}_{\text{ind}} \to B^{(n)}_{\text{ind}} \to H^{(n+1)}. \tag{29}$$

The second scheme should be attempted only after the induced Zeeman correction is numerically stable.

Optional extension: on-site Hubbard magnetism. If the spin polarization produced by the induced Zeeman splitting alone is too weak, a local Hubbard term can be added,

$$\hat{H}_U = U \sum_l \hat{n}_{l\uparrow} \hat{n}_{l\downarrow}. \tag{30}$$

At mean-field level, this gives a spin-dependent on-site correction

$$H^{\text{MF}}_U = U \sum_l \left(\langle n_{l\downarrow} \rangle \hat{n}_{l\uparrow} + \langle n_{l\uparrow} \rangle \hat{n}_{l\downarrow} \right), \tag{31}$$

so that the effective on-site potentials become

$$V_{l\uparrow}^{\text{eff}} = e\phi_l + U\langle n_{l\downarrow}\rangle + \mu_B B_{\text{ind},z}(\mathbf{r}_l, t), \qquad V_{l\downarrow}^{\text{eff}} = e\phi_l + U\langle n_{l\uparrow}\rangle - \mu_B B_{\text{ind},z}(\mathbf{r}_l, t). \tag{32}$$

This should be treated as an optional extension beyond the main scope.[5]

10 Final Goals

First goal: A quantitative phase diagram for when induced magnetic effects matter in the nonlinear optical response of graphene islands. That means, for several island sizes and edge terminations, we should map how the three levels of implementation changes,

- the change in linear extinction
- the change in harmonic yield

We can do this with the kind of doping maps of different geometries (zigzag vs armchair, triangle vs. hexagon vs. bow-tie) and sizes already produced. For the change, try to plot the relative correction rather that the difference,

$$\delta_{mag} = \frac{\sigma_J - \sigma_0}{\sigma_0}$$

Second goal: Investigate the hypothesis that plasmon-enhanced optical excitation in graphene islands produces a spin asymmetry in the current response, strongest near resonant frequencies and in specific geometries.

We can do this by mapping the spin current in islands with a significantly induced Zeeman splitting. We should calculate

$$J_\alpha = J_\alpha^\uparrow + J_\alpha^\downarrow,$$

and

$$J_\alpha^s = J_\alpha^\uparrow - J_\alpha^\downarrow.$$

where $\alpha = \{x, y\}$ is the field direction. You can get the frequency dependent current by Fourrier transformation. Then plot $|J_\alpha(\omega)|^2$, $|J_\alpha^s(\omega)|^2$, J_α^s, $\delta J_\alpha(L0 \to L1)$, and $\delta J_\alpha(L1 \to L2)$ off- and on resonance. Alternatively create a output frequency vs doping contour plot of the quantities.

We should create results that demonstrate,

- Induced spin-resolved currents vs. frequency in linear response,
- Induced spin-resolved currents vs. frequency for a Gaussian pulse.
- Real-space induced current at the resonance frequency for systems with large magnetic response.

We hope to see a stronger effect at the plasmon resonances and for zigzag edge terminations.

11 Note on gauge invariance

Gauge invariance means that physical observables must be unchanged under

$$\mathbf{A} \to \mathbf{A} + \nabla\chi, \qquad \phi \to \phi - \partial_t\chi, \qquad |l\rangle \to e^{-ie\chi(\mathbf{r}_l,t)/\hbar}\,|l\rangle\,. \tag{33}$$

In the present project the external optical field is kept entirely in the length gauge through $\phi_l^{\text{ext}}(t) = -\mathbf{r}_l \cdot \mathbf{E}_{\text{ext}}(t)$, and there is no external vector potential. Therefore the electrostatic implementation does not require any comparison between different external magnetic gauges. After the induced vector potential A_{ind} is added, a gauge-related sensitivity test does become necessary. This is described explicitly in the Biot–Savart section above.

Gauge invariance of phases. The phase on a single bond is gauge dependent, but the sum around a closed loop is physical,

$$\sum_{\langle ll' \rangle \in hex} \theta_{ll'}^{\text{ind}} = \frac{e}{\hbar} \oint_{hex} \mathbf{A}_{\text{ind}} \cdot d\mathbf{r} = \frac{e}{\hbar} \Phi_{hex}^{\text{ind}}. \tag{34}$$

Thus the induced magnetic flux through a carbon hexagon, not the individual bond phase, is the physical quantity.

References

[1] Line Jelver and Joel D. Cox. Nonlinear plasmonics in nanostructured phosphorene. *ACS Nano*, 17:20043–20052, 2023.

[2] Joel D. Cox and F. Javier García de Abajo. Electrically tunable nonlinear plasmonics in graphene nanoislands. *Nature Communications*, 5:5725, 2014.

[3] Joel D. Cox, Ivan Silveiro, and F. Javier García de Abajo. Quantum effects in the nonlinear response of graphene plasmons. *ACS Nano*, 10(2):1995–2003, 2016.

[4] M. Graf and P. Vogl. Electromagnetic fields and dielectric response in empirical tight-binding theory. *Physical Review B*, 51:4940–4949, 1995.

[5] David Jacob and Joaquín Fernández-Rossier. Theory of intermolecular exchange in coupled spin-1/2 nanographenes. *Physical Review B*, 106:205405, 2022.